

Lecture - 9

Analytical system administration

Section Overview

- ◆ Why monitoring?
- ◆ Historical monitoring
- ◆ Real-time monitoring
- ◆ Monitoring techniques
- ◆ Monit
- ◆ Performance monitoring.
- ◆ Performance tuning.
- ◆ Automation of Periodic Tasks
- ◆ Scheduling and Cron
- ◆ Syslog
- ◆ Accounting

Why Monitoring?

“If you aren’t monitoring a service,
you can’t manage it.”

Why Monitoring?

1. Rapidly detect and fix problems.
2. Identify the source of problems.
3. Predict and avoid future problems.
4. Document an SA's achievements.

Historical Monitoring

Record long-term system statistics.

- Uptime.

- Performance.

- Security.

- Utilizations.

Examples

- Web server uptime was 99.99% last year, compared to 99.9% the previous year.

- Peak network usage is 8 MBps, up from 5 MBps last year.

Uses

- Capacity planning.

- Planning for reliability or security improvements.

Historical Monitoring Processes

Polling

- Take measurements at regular intervals.

- Store database of measurements.

- Graph summaries of collected data.

Measurement Tools

- `iostat`

- `vmstat`

- `ps`

- `sar`

Real-time Monitoring

Alert SA to failures as they happen.

Discover problems before customer does.

Shorter outages.

Better reputation.

Real-time Monitor components

Monitoring system (poll or alert).

Notification system.

Real-time Monitoring Techniques

Polling

Poll systems and applications for status.

Ex: ping critical servers every 5 minutes.

Alerting

Many systems can send alerts to monitoring system when they detect a problem.

Ex: RAID array logs a disk failure.

Notification

Types of notification

1. Email
2. Paging
3. Phone call

Reliability

1. Notification system should not depend on system being monitored.
2. Email can fail or have long delays.
3. Pages are susceptible to third party failures and monitoring.

Escalation

What if the SA is on vacation?

Notifications need to be transferrable.

Static: reconfigure notifier before vacation.

Dynamic: configurable set of recipients.

Ex: If SA doesn't respond in 1 hour, notify manager.

Types of monitoring

Availability

Watch for outages in network, host, apps.

Ex: cannot reach mail server.

Capacity

Check thresholds for CPU, mem, disk, network.

Ex: mail spool disk is 95% full

Active Monitoring

Active monitoring systems can fix problems.

1. Respond faster than a human can.
2. Can typically only implement temporary fix.
3. Can't fix some problems: bad disk, out of paper.

Risks

Reliability: Test active responses thoroughly before deployment.

Security: Active monitor typically needs admin access on all monitored systems.

Levels of Testing

1. Check server is pingable.

Verifies network connectivity from monitor only.

2. Check that application is up.

Make a TCP connection to service port.

Check process or service list.

3. End-to-end testing.

Entire transaction as customer would do.

Ex: send and receive an e-mail message.

Running `monit`

Starting

```
monit [-v]
```

Status

```
monit status
```

```
monit summary
```

(also provides web interface on port 2812)

Stopping

```
monit quit
```

Global configuration

```
set daemon 60
set logfile syslog facility log_daemon
set alert root@domain
set httpd port 2812 address localhost
    allow localhost
    allow admin:monit
```

Monitoring a Process

```
check process apache
  with pidfile "/usr/local/apache/logs/httpd.pid"
  start = "/etc/init.d/httpd start"
  stop = "/etc/init.d/httpd stop"
  if failed port 80 and protocol http
    and request "/cgi-bin/printenv"
    then restart
  if cpu usage is greater than 60 percent for
    2 cycles then alert
  if cpu usage > 98% for 5 cycles then restart
  if 2 restarts within 3 cycles then timeout
```

Monitoring a File

```
# Rotate log if it gets too big
check file access_log
    with path /var/log/access_log
    if size > 100 Mb
        then exec "/usr/sbin/logrotate -f
rotate_apache_now"
# Restart Apache if config changes
check file httpd.conf
    with path /usr/local/apache/conf/httpd.conf
    if changed checksum then exec
        "/usr/local/apache/bin/apachectl
graceful"
```

Monitoring CPU

```
check system localhost
```

```
if loadavg (1min) > 5 then alert
```

```
if loadavg (5min) > 3 then alert
```

```
if memory usage > 80% then alert
```

```
if cpu usage (user) > 80% then alert
```

Monitoring a Disk

```
check device rootfs with path /  
    if space usage > 90% then alert  
check device varfs with path /var  
    if space usage > 90% then alert
```

Monitoring Remote Hosts

```
# Ping the host to see if it's up
check host foo with address foo.com
    if failed icmp type echo
        with timeout 15 seconds then alert
# Detailed test, accessing web services
check host foo with address foo
    if failed port 80 protocol http and
        request "/status" then alert
    if failed port 443 type TCPSSL and
        protocol http with timeout 15
        seconds then alert
```

Automation and Observation

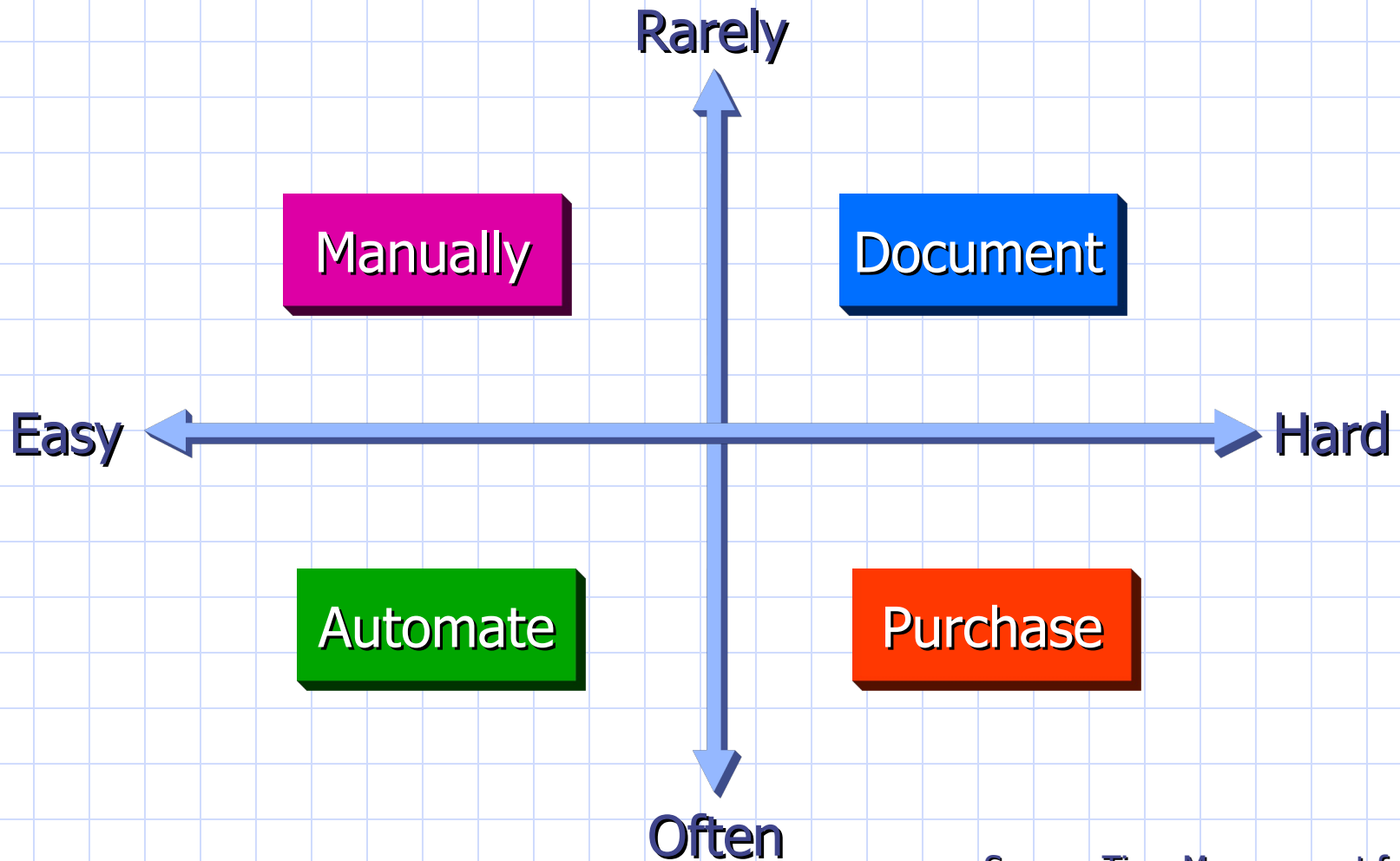
◆ Automation

- Simplify repetitive tasks
- Shell Scripting
- Task Scheduling

◆ Observation

- Current
- Historical

SA Task Classification



Source: Time Management for SAs
Thomas A Limoncelli

Periodic Processes

◆ Some tasks need to be run at set times

◆ **crond**

- Runs programs specified in a **crontab** file
- Each user has own **crontab** file
- **crontab** command used to modify **crontab** files

crontab File Format

Field	Description	Range
Minute	Minute of the hour	0 – 59
Hour	Hour of the day	0 – 23
Day	Day of the month	1 – 31
Month	Month of the year	1 – 12
Weekday	Day of week (Sun – Sat)	0 – 6
Command	Command to run	

crontab Range Format

Format	Description
Value	Exact value
*	Match all values
Val1 – Val2	Match values between Val1 and Val2
V1 – V2 / Step	Every <step> between V1 and V2
Val1,Val2	Match Val1 and Val2

Note: '-', '/', and ',' can be combined

crontab Command

crontab [-e|-l|-r] [user]

- ◆ **-e**: Edit the crontab file
- ◆ **-l**: List the contents of the crontab file
- ◆ **-r**: Remove the crontab file
- ◆ Root can specify other user crontabs

Access Control for `cron`

- ◆ Can control which users may use `cron`

- ◆ `cron.allow`

- List of users permitted to use `cron`
- Checked first

- ◆ `cron.deny`

- List of users denied access to `cron`
- Checked if `cron.allow` does not exist

ISC ("Vixie") Cron

- ◆ Replacement for standard cron daemon
- ◆ `/etc/crontab` – System crontab file
 - Inserted "run-as" field (6)
 - `run-parts`
- ◆ Scripts put into `/etc/cron.<period>`
 - `cron.hourly` (run 1 minute after every hour)
 - `cron.daily` (run 4:02 daily)
 - `cron.weekly` (run 4:22 every Sunday)
 - `cron.monthly` (run 4:42 first of every month)

Windows Task Scheduler

◆ Administrative Tools->Task Scheduler

◆ Trigger based

- Time
- State

◆ Special Conditions

- Idle
- Power (AC or Battery)
- Network Connections

◆ Actions

- Run a program
- Send email
- Display a message

Current System Status

◆ Disk Space Usage

- `du`: Disk space used by file/directory
- `df`: Disk space used by file system

◆ Memory/CPU Usage

- `ps`
- `uptime`
- `free/swap`
- `top/System Monitor`

Historical Observation

◆ Log files

- Server daemons
- RSyslog
- Automated tools
 - ◆ swatch
 - ◆ Logcheck
 - ◆ Splunk

◆ Accounting

- Logins/logouts
- Process usage

◆ `/var/log`

rsyslog: System Logger

- ◆ Central logging facility

- ◆ Standard API

- ◆ Components

- `syslogd`

- `/etc/rsyslog.conf`

- `logger`

- Log files

rsyslog.conf

Format: **selector** **action**

◆ Selector – facility.level

- Facility - Who or What
 - ◆ ';' – Separates multiple facilities
- Level – When
 - ◆ '=' – Matches level
 - ◆ No '=' – Matches level and all above
- ';' – Separates multiple selectors
- '*' – Match all facilities or levels

◆ Action - What to do with the message

Common Syslog Facilities

Facility	Source
kern	Kernel
mail	Sendmail
lpr	Printing
daemon	System Daemons
cron	Cron Daemon
user	User processes (default)
local0-7	Locally assigned
auth	Security & Authentication

Syslog Severity Levels

Level	Condition
emerg	Panic situations
alert	Urgent situations
crit	Critical conditions
err	Other error conditions
warning	Warning messages
notice	Things to check?
info	Information messages
debug	Debugging only

Syslog Actions

Action	Meaning
<i>filename</i>	Writes message in <i>filename</i>
<i>@hostname</i>	Forwards message to <i>hostname</i>
<i>@ipaddress</i>	Forwards message to host at <i>IP</i>
<i>user1,user2,...</i>	Send to user screens (if logged in)
<i>*</i>	Send to all logged in user screens

Multiple actions require multiple lines!

Login Accounting

◆ **wtmp** – DB of all logins and logouts

- Time
- User/TTY
- Where

◆ **utmp** – DB of currently logged in users

◆ **Reports**

- **who/w** – Lists currently logged in users
- **last** – Lists all login sessions
- **lastlog** – List last time users logged in

Process Accounting

◆ Process information tracked

- Users
- Commands run
- CPU, memory, and I/O usage

◆ Accounting system

- `accton` – Turns accounting on
- `lastcomm` – last command run by user
- Vendor specific tools

◆ Can eat a lot of disk space quickly!!!

Windows Event Viewer

◆ Administrative Tools->Event Viewer

◆ Event Logs

- Windows Logs

- ◆ Application
- ◆ Setup
- ◆ Security
- ◆ System
- ◆ Forwarded

- Application and Service Logs

◆ Subscriptions

Windows Event Levels

- ◆ Critical
- ◆ Error
- ◆ Warning
- ◆ Information
- ◆ Verbose
- ◆ Audit (Security)
 - Success
 - Failure

Why track usage?

- ◆ \$\$\$ - Charging for usage
- ◆ Track user abuse of resources
- ◆ Generating a usage baseline for users
- ◆ Reports for management

How long to keep logs?

- ◆ Don't log at all
- ◆ Reset the logs periodically
- ◆ Rotate log files
 - Via cron
 - `date` command
- ◆ Permanently archive log data
 - File compression tools
 - Tape
 - CDROM